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On the Problem of Polymorphism and Fusion of Lanthanide Trifluorides. I. The Influence of Oxygen on Phase Transition Temperatures*

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The details of using thermal analysis (TA) in the investigation of lanthanide trifluorides are examined. By examining the literature on the problem of phase transitions in LnF_3 (Ln = lanthanum and the lanthanides), it is established that the basic reason for disagreement among the data of various authors is, in most cases, lack of control of pyrohydrolysis of LnF_3.

For the first time, parts of the phase diagrams of the LnF_3–Ln_2O_3 systems (Ln = Gd, Tb, Ho, Er, and Y) have been studied with TA methods. Phases $LnF_{3-2x}O_x$ where $0 \leqslant x \leqslant 0.2$, which were not known earlier, are discovered. According to the type of interaction of LnF_3 with the corresponding Ln_2O_3, the trifluoride series is broken down into several groups corresponding to the structural type in which the LnF_3 crystallizes. The parts of the phase diagrams of the LnF_3–Ln_2O_3 systems that were studied permit us to explain the reasons for contradictions in existing data on the temperatures of phase transformations in LnF_3.

Introduction

A large number of investigations have been dedicated to the study of polymorphism and fusion of the trifluorides of the rare-earth elements, the basic conclusions of which we summarize here. Most authors have obtained concurring data on the fusion temperatures of (Y, Ln)F_3 (Ln = lanthanum and the lanthanides). Disagreement in the values of the temperatures of polymorphic transitions and conditions of morphotropic transitions as cited by the various authors clearly goes beyond the limits of possible error in the experiments. For a critical analysis of the information available in the literature, we used not only the results of a study of pure (Y, Ln)F_3 but also the results of our investigation of the binary systems LnF_3–Ln_2O_3 and LnF_3–MF_2 (M = Ca, Sr, Ba; Ln = lanthanum and the lanthanides).

History of the problem. The first indication of two forms (structure types tysonite LaF_3 and orthorhombic β-YF_3) in which several trifluorides of lanthanides and yttrium crystallize is contained in (*1*). Later these data were confirmed several times. A systematic investigation with a high-temperature X-ray method (*2*) of the polymorphism of the entire (Y, Ln)F_3 series revealed, in addition to the two already known crystal forms, a third previously unknown one, called by the authors a type of "hexagonal YF_3". According to (*2*), lanthanide trifluorides from Er to Lu and YF_3 crystallize in this type. Quenching of this high-temperature form was not possible. In (*3*), the "hexagonal YF_3" type is identified (by powder X rays) as α-UO_3 type. The authors of (*2*) were the first to present a general diagram of the phase transformations in the LnF_3 series, as indicated in Fig. 1 (curves 1 and 2), using fusion temperatures of LnF_3 as cited in (*4*). According to (*2*), all the (Y, Ln)F_3 with Ln = Sm–Lu are dimorphic; the morphotropic change in the (Y, Ln)F_3 high-temperature

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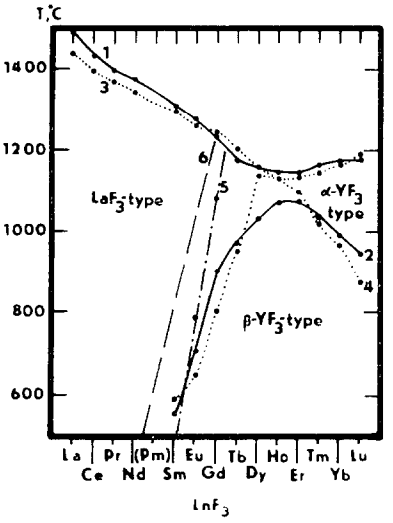


FIG. 1. Data in the literature on phase transition temperatures in the lanthanide trifluorides (explanation in the text).

modifications from tysonite structure type (Ln = La–Ho) to α-YF_3 type (Ln = Er–Lu and Y) coincides with the minimum on the fusion curve 1 and the maximum on the orthorhombic modification thermal stability curve 2 (Fig. 1).

The results of all the systematic studies of the phase transition of LnF_3 published before 1975 are presented in Fig. 1. The phase transformation temperatures are shown in Table I. For the fusion temperatures in Fig. 1, only the data of (*4*) (curve 1) and (*5*) (curve 3) are presented as being the most systematic and, in addition, the most noticeably divergent.

The authors of (*5*) have presented another diagram of the polymorphism of LnF_3 (curves 3 and 4, Fig. 1). According to (*5*), HoF_3 crystallizes from the melt in an orthorhombic form, and the temperatures of the phase transformation $\alpha \rightleftarrows \beta$-$DyF_3$ and fusion almost coincide. This diagram is based on growth of monocrystals of orthorhombic DyF_3 and HoF_3 out of the melt.

The series of studies (*6–11*) on the temperatures of the $\alpha \rightleftarrows \beta$-$LnF_3$ transformation, where Ln = Eu-Dy leads to a third polymorphism diagram (curve 5, Fig. 1), accord-

TABLE I

LITERATURE DATA ON PHASE TRANSITION TEMPERATURES IN TRIFLUORIDES OF YTTRIUM, LANTHANUM, AND THE LANTHANIDES

RF_3	Fusion temperature (°C)	Polymorphic transformation temperature (°C)
YF_3	1145 [5][a]; 1155 [10, 30]; 1148 [24]; 1149 [25]; 1143 [26]; 1135 [31]; 1144 [29]	1052 [2, 29]; 1050 [5]; 1077 [10]; 1045 [25]; 1040 [26]; 1080 [30]; 1065 [31]
LaF_3	1445 [5]; 1493 [4, 10]; 1490 [24]; 1487 [25]; 1495 [34]	—
CeF_3	1430 [4]; 1400 [5]; 1432 [11]; 1437 [24]; 1425 [34]	—
PrF_3	1395 [4]; 1368 [5]; 1399 [10]; 1400 [34]	—
NdF_3	1374 [4]; 1342 [5]; 1377 [10]; 1360 [26]; 1380 [34]	—
SmF_3	1306 [4]; 1295 [5]; 1310 [7]; 1300 [11]	555 [2]; 580 [5]; 490 [11]
EuF_3	1276 [4]; 1262 [5, 11]	700 [2]; 780 [8, 33]; 647 [11]
GdF_3	1231 [4]; 1239 [5]; 1205 [6]; 1250 [9]; 1230 [10,13]; 1232 [11]; 1228 [24]; 1225 [32]	900 [2, 13]; 805 [5]; 1077 [4]; 1060 [6]; 1075 [9, 11]; 1074 [10]; 1100 [13]; 1066 [32]
TbF_3	1172 [4]; 1200 [5]; 1177 [11]	950 [2]; 962 [5]
DyF_3	1154 [4, 5]; 1108 [7]; 1157 [11]; 1160 [13]	1030 [2]; 1140 [5]
HoF_3	1143 [4, 10]; 1130 [5]; 1150 [13]; 1145 [35]	1070 [2, 10]; 1130 [5]; 1125 [13]; 1075 [13]; 1070 [35]
ErF_3	1140 [4, 14]; 1132 [5]; 1146 [11]; 1145 [27]	1075 [2]; 1096 [5]; 1117 [11]; 1096 [27]
TmF_3	1158 [4, 11]; 1143 [5]	1030 [2]; 1018 [5]; 1053 [11]
YbF_3	1157 [4]; 1168 [5]; 1162 [11]; 1145 [27]; 1165 [28]	985 [2]; 971 [5]; 986 [11]; 1000 [27]; 978 [28]
LuF_3	1182 [4, 5]; 1184 [10]	945 [2]; 870 [5]; 957 [10]

[a] Temperature values taken from the graph of Ref. (*5*).

ing to which the last dimorphic LnF_3 with a high-temperature modification of the LaF_3 is gadolinium fluoride. It will be shown below that our data also coincide with this diagram.

The fourth diagram (curve 6, Fig. 1) is the result of a series of studies (*12–14*) on the growth of LnF_3 single crystals after extended treatment of the superheated melt with a stream of anhydrous HF, which had been used even earlier by (*15–19*) and others for thorough removal of oxygen. The authors of (*12–14*) succeeded in obtaining crystals of orthorhombic trifluoride of Gd, Tb, Dy, Ho, Er, and Y from the melt. Unfortunately, these studies do not give quantitative analyses of the starting fluorides and the melts after treatment with HF, which makes interpretation of the results difficult.

Thus, a brief survey of published data on the polymorphism of LnF_3 indicates that the diagram suggested by (*2*) has been basically transformed over the past years. New experimental data have turned out to be entirely different from one another; as a result, at the present time, the problem of polymorphism and morphotropism in the LnF_3 series cannot be considered solved.

A clear definition of the great influence of partial substitution of fluorine by oxygen in LnF_3 on the phase transformation temperatures has not been investigated experimentally for the majority of LnF_3.

The aim of the present work is to study the influence of admixture of oxygen on the fusion and polymorphic transformation temperatures and also regularity of change of crystal-structure type in the LnF_3 series (morphotropism). The work consists of two parts. In the first, the results of a study of the parts of the phase diagrams of LnF_3–Ln_2O_3 in the vicinity of the rare earth trifluoride are presented. In the second, the conditions of thermal stabilization of the LnF_3 structure types by heterovalent isomorphous substitution of Ln^{3+} by M^{2+} in the LnF_3–MF_2 (M = Ca, Sr, Ba) binary systems are looked into. From the study of the LnF_3–Ln_2O_3 and LnF_3–MF_2 systems, conclusions are drawn about the location of morphotropic transitions in the LnF_3 series and a diagram of the polymorphism of LnF_3 is presented, reflecting the current state of their study and the level attained in their purification. Finally, values for phase transformation temperatures in LnF_3 at determined levels of oxygen content are presented.[1]

Experimental

Materials. Trifluorides of yttrium, lanthanum, and the lanthanides obtained from oxides and fused in fluorinating atmosphere were used. Oxygen was determined by the vacuum-fusion method (*17*) with a sensitivity of 0.001 wt% and a relative error of ±25%.

Method. Phase transformations in LnF_3 were examined by thermal analysis (TA) in a set-up with a ceramicless heater described earlier (*18*). The analysis was carried out in nickel or molybdenum thin-walled crucibles with pockets for W5% Re/W20% Re thermocouples inserted from below. The mass of the specimen was from 1 to 2.5 g.

The closed chamber of the TA set-up was filled with pure helium with additional absorption of any traces of oxygen by heated titanium sponge and the molybdenum. The method gives oxygen content in the specimens after analysis on an average of 0.04 wt% (the oxygen content for each LnF_3 will be given in the second part).

For the temperature range 1000–1500°C, the calibration of thermocouples was made difficult by absence of standard benchmark materials that do not react with the crucible materials. Fluorides, for which closely agreeing data on fusion temperatures have been produced in the literature, i.e., CaF_2, 1418°C (*16*, *19–22*), and YF_3, 1152°C (see Table I), were used. Fusion temperatures of other fluorides referred to such "standard materials" agree well with those reported earlier: SrF_2, 1464°C; BaF_2, 1355°C. A comparison of the fusion temperatures of LnF_3 with the data of (*10*, *11*) is given in the second part. The accuracy of the temperature determination is estimated to be on the average ±10°C; reproducibility amounts to ±3°C.

[1] We became acquainted with Ref. (*11*) when the larger part of the published data on phase transition temperatures had already been received. We confirm with satisfaction the close accord of our determinations with the data in (*11*).

A record of the cooling curves of LnF_3 discloses a clear tendency of their melts to supercool, which had been noted for individual trifluorides even earlier (see, for example *11*). On the heating curves, the temperature registered for the beginning of the fusion is independent of heating rate within the range 4–20°/min.

In our study of polymorphic transitions, the rates of heating or cooling had an effect on the observed transformation temperatures, as mentioned earlier (*23*). Figure 2 shows the dependence of measured temperature of $\alpha \rightleftarrows \beta$ transformations for YF_3 and GdF_3 on heating (cooling) rate. It is seen that the temperature hysteresis is extremely large for both trifluorides. Therefore, in determining the polymorphic transition temperatures for LnF_3 (as well as fusion temperatures), it is necessary to use only the heating curves. Averaging the temperatures obtained on heating and cooling, as had been done, for example, by (*5*, *26*), leads to large errors.

The form of the dependence of transformation temperature on cooling rate allows us to propose that these curves have a horizontal asymptote, i.e., it is impossible to supercool the high temperature forms of α-YF_3 and α-GdF_3 below a set temperature (spinodal).

Two-component mixtures of the LnF_3–Ln_2O_3 systems were homogenized either ahead of time or directly in the TA set-up. The time maintained at fusion was determined experimentally and came to 1.5–3 hr.

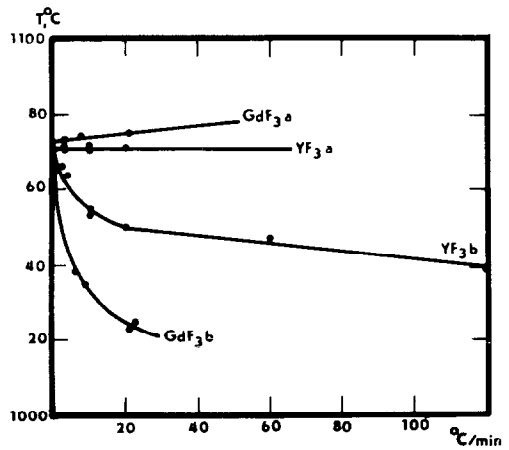


FIG. 2. Dependence on the rates of (a) heating and (b) cooling of measured temperatures of $\alpha \rightleftarrows \beta$ transformations for GdF_3 (0.005 wt% oxygen) and YF_3 (0.010–0.045 wt% oxygen).

X-ray analysis was carried out on a AFV-202E (Japan) diffractometer–spectrometer, which was equipped with a high-temperature attachment (heater of platinum–rhodium wire, sapphire insulation, molybdenum thermal screens).

Results and Discussion

1. Effect of Oxygen on the Fusion Temperatures of Lanthanide Trifluorides

There are few data on the initial stages of fluorine substitution in LnF_3 by oxygen. The first intermediate products are thought to be oxyfluorides $LnF_{3-2x}O_x$ where $x \geqslant 0.7$ (see, for example, *36–38*). Only in (*33*), where the formation of a solid solution of europium oxide (3+) in a tysonite modification of EuF_3 was observed, is a state diagram scheme of the EuF_3–Eu_2O_3 system shown. The solid solution has its maximum on the melting curve and breaks up by the eutectoid scheme. In (*32*), it was found that the first phase in the hydrolysis of EuF_3 is the ternary compound Eu_2OF_4 ($x = 0.5$); unfortunately, it has not been studied in detail.

Qualitative indication of a rise in the fusion temperature of YF_3 by contamination with oxygen is shown in (*31*). According to (*10*, *11*), admixture of oxygen to the extent of 0.005–0.075 wt% did not show a noticeable effect on the melting temperature of LnF_3; (*13*) shares this view for HoF_3 and GdF_3. The authors of (*6*) tell us that the addition of 2% Gd_2O_3 raises the fusion temperature of GdF_3 by 22°C. According to (*40*), LaF_3 forms a eutectic with La_2O_3 without any anomaly on the melting curve near LaF_3.

Systematic data about the initial stages of interaction of LnF_3 with Ln_2O_3, as indicated in the foregoing survey, are lacking. The effect of oxygen on the phase transition temperatures in LnF_3 is not always the same: The substitution of fluorine by oxygen can raise as well as lower the fusion temperature of different LnF_3. [A statement about the generality of the action of all admixtures on LnF_3, raising

fusion temperatures, is contained, for example, in the recent (*13*)].

We have studied several LnF_3–Ln_2O_3 systems. The choice of LnF_3 was taken from those lanthanide trifluoride series where, according to data in the literature (see Fig. 1), one could expect a change of structural type.

Figure 3 (A–E) shows parts of the phase diagrams of LnF_3–Ln_2O_3 systems with Ln_2O_3 contents up to 8 mole%. [In the case of B, the terbium oxide $Tb_2O_{3.5}$ (Tb_4O_7) was used.] To explain the nature of the thermal effects we have used construction of the Tamman triangle. It can be seen that in the systems studied the effect of admixture of oxygen on the fusion temperature becomes noticeable when the oxygen content in $LnF_{3-2x}O_x$ is $x \leq 0.01$–0.02. This corresponds closely to an oxygen content of 0.10–0.15 wt% and confirms the conclusion of (*10*, *11*) as to the practical insensitivity of fusion temperatures of LnF_3 to oxygen admixture of less than 0.07 wt%.

In Fig. 3 it is seen that in all cases the interaction of LnF_3 with Ln_2O_3 is accompanied by formation of solid solutions of $LnF_{3-2x}O_x$ (in Fig. 3 the areas of homogeneity are shaded). In none of the systems studied did we succeed in quenching these solid solutions (in the powder X-rays, reflections of orthorhombic LnF_3 and tetragonal oxyfluoride are present). However, the formation of $LnF_{3-2x}O_x$ phases is clearly registered by the TA method. As can be seen from Fig. 3, their position on the phase diagrams changes depending on the lanthanide.

In GdF_3–Gd_2O_3 (Fig. 3A), the solid solution $GdF_{3-2x}O_x$ is formed based on tysonite form α-GdF_3. The dissolution of Gd_3O_2 into GdF_3 is accompanied by a significant rise in the fusion temperature of the $GdF_{3-2x}O_x$ phase. The maximum on the fusibility curve exceeds the fusion temperature of GdF_3 by 90°C. The data of different authors for the fusion temperature of GdF_3 (see Table 1) fall into a wide range: from 1205 (*6*) to 1250°C (*9*). This agrees with the fact that in the case of dimorphic trifluorides from SmF_3 to GdF_3, admixture of oxygen clearly shows a pronounced thermal stabilization of the high-temperature modification of tysonite structural type.

As is evident from Figs. 3B and C, the interaction of orthorhombic LnF_3 with the corresponding oxides also leads to formation of solid solutions $LnF_{3-2x}O_x$. However, these phases are topologically different from the $GdF_{3-2x}O_x$ described above. They are separated from pure LnF_3 by two-phase areas and form eutectics with them. The fusion tempera-

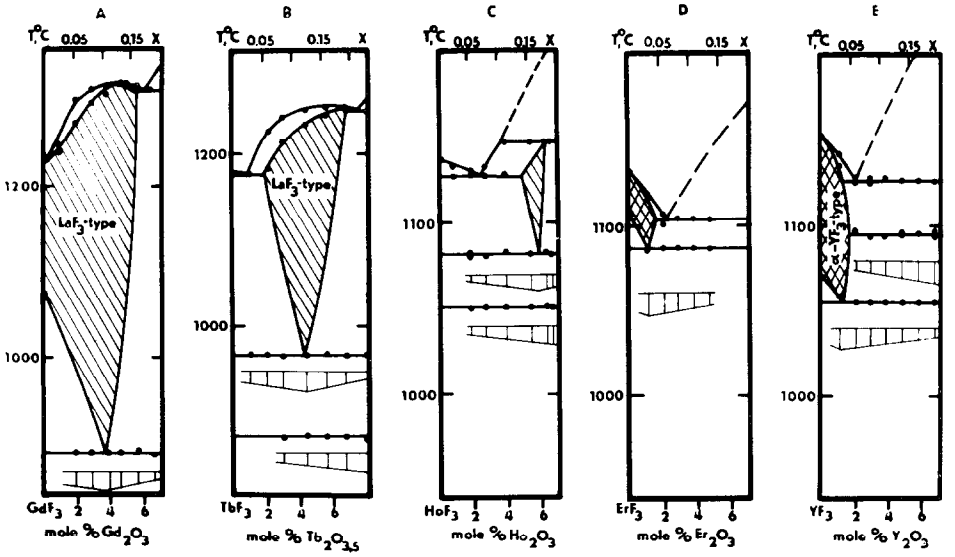


FIG. 3. Parts of the phase diagrams of several LnF_3–Ln_2O_3 systems.

tures of eutectics are extremely close to the fusion of LnF_3; therefore, small oxygen contents practically do not change the fusion temperature of compositions in the LnF_3–Ln_2O_3 systems with orthorhombic trifluorides. Further increase in the concentration of Ln_2O_3 for this LnF_3 group (Ln = Tb–Ho) leads to different behavior of the $LnF_{3-2x}O_x$ phases. In the system with TbF_3, this phase melts congruently; in the system with HoF_3, it melts incongruently.

In the group of LnF_3 that have a high-temperature modification of the type α-YF_3, formation of solid solutions $LnF_{3-2x}O_x$ based on it (Figs. 3D and E) is also observed. In both systems studied (the area of homogeneity of the $LnF_{3-2x}O_x$ phases of the α-YF_3 type is shown in Fig. 3 by double hatching), the dissolution of Ln_2O_3 into α-LnF_3 is accompanied at first by noticeable decrease in the fusion temperature. After reaching the eutectic composition, further increase of the oxygen content is apparently accompanied by a sharp rise in fusion temperature.

Thus, the results of studying parts of the phase diagrams of five of the LnF_3–Ln_2O_3 systems showed that the interaction of LnF_3 with an admixture of oxygen certainly does not have a universal character. The division of LnF_3 series into subgroups according to polymorphic and morphotropic transitions reflects the structural types and the position of the $LnF_{3-2x}O_x$ phases on the phase diagrams.

The mechanism of charge compensation during the heterovalent substitution of fluorine by oxygen in LnF_3 is difficult to establish, since it is impossible to measure the density of the $LnF_{3-2x}O_x$ phases because of their failure to quench. Measurements on the density of $BiF_{3-2x}O_x$ phases with tysonite structure that are formed during pyrohydrolysis of BiF_3 (*41*) give indirect evidence in favor of a vacancy mechanism. The low energy of activation of the vacancies in LaF_3 obtained by (*42*) also testifies to appearance of tysonite high-temperature modifications in LnF_3 where Ln = Sm–Gd. Apparently, the dimorphism of this LnF_3 group can be viewed as consequence of the stabilization of the tysonite structure by vacancies produced by the temperature rise.

The great tendency of LnF_3 toward pyrohydrolysis and the data we have obtained for trifluorides from Gd to Ho inclusive on formation of new phases $LnF_{3-2x}O_x$ ($0 \leqslant x \leqslant 0.2$) not previously described leads to the conclusion that the tysonite forms of LnF_3, Ln = Tb–Ho, observed by a number of authors are, apparently, solid solutions of $LnF_{3-2x}O_x$ (see below).

2. *The Effect of Oxygen on the Temperature of Polymorphic Transformations of Lanthanide Trifluorides*

The effect of oxygen on the temperature of polymorphic transformation of LnF_3 can be seen in Figs. 3A–E. The solid solutions that are formed decompose according to a eutectoid scheme in all the systems, independent of the type of structure (LaF_3 or α-YF_3). Therefore, in those cases (Figs. 3A, D, and E) where solid solutions $LnF_{3-2x}O_x$ are formed based on a high-temperature modification of LnF_3, one can speak about the universal effect of oxygen in raising the temperature of the polymorphic transformation of LnF_3. In systems with terbium and holmium (Figs. 3B and C), the $LnF_{3-2x}O_x$ phases are separated from LnF_3 by two-phase regions and have berthollide character. For this LnF_3 group, no dissolution of rare earth element oxide into orthorhombic LnF_3 is observed. Our data on the growth of HoF_3 single crystals indicate that the maximum solubility of oxygen in orthorhombic HoF_3 does not exceed 0.005 wt%.

We have not studied the structure of the $LnF_{3-2x}O_x$ phases (Ln = Gd–Ho) directly. X-ray study at high temperatures (*2*) can serve as indirect evidence that they belong to the tysonite structural type. Agreement within the limits of the accuracy of the experiment between our data on the temperatures of the eutectoid decomposition of $LnF_{3-2x}O_x$ and the "transition" temperatures of LnF_3 into the tysonite form (*2*) is seen from Fig. 4. It can be concluded that the authors of (*2*), in the case of most of the LnF_3 really observed products of a partial hydrolysis of LnF_3. According to their determinations for Ln = Gd–Ho, these products at high temperature had a tysonite structure that did not quench. The close proximity of the $LnF_{3-2x}O_x$ phases

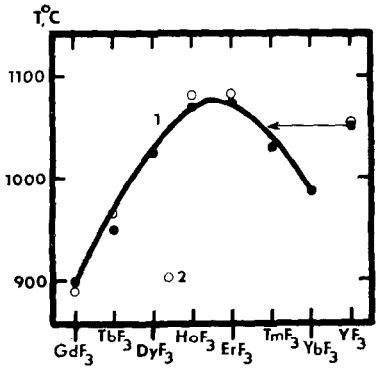


FIG. 4. Curve (1) of "polymorphic transitions" in LnF_3 according to (*2*) compared with eutectoid decomposition temperatures (points 2) of the $LnF_{3-2x}O_x$ phases ($0 \leq x \leq 0.2$) according to our data.

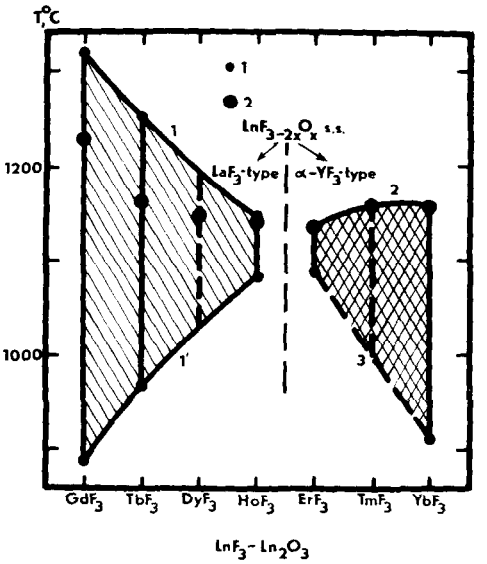


FIG. 5. Temperature stability zones of $LnF_{3-2x}O_x$ phases in the LnF_3–Ln_2O_3 systems (points 1). Points (2) are the fusion temperatures of LnF_3.

to the LnF_3 ordinates led to the fact that they erroneously took the high temperature modifications of LnF_3 (Ln = Tb–Ho) for tysonite.

From a comparison of parts of the phase diagrams of the LnF_3–Ln_2O_3 systems (Figs. 3A–C), it is seen that the temperature range for the equilibrium existence of $LnF_{3-2x}O_x$ phases for Ln = Gd–Ho is regularly related to the increase in the atomic number of the lanthanide. This dependence is presented graphically in Fig. 5. Extrapolation of curves 1,1′ (fusion and decomposition temperatures of $LnF_{3-2x}O_x$, respectively) indicates that the tysonite structure of the $LnF_{3-2x}O_x$ phase may evidently not be seen in the tysonites of lanthanides "heavier" than holmium. Actually, the high-temperature X-ray studies of (*2*) showed that the partially hydrolyzed trifluorides from erbium to lutetium and yttrium crystallize in the other structural type (α-YF_3). We have observed partial hydrolysis of YbF_3 in the high-temperature X-ray camera. A $YbF_{3-2x}O_x$ phase with the α-YF_3 structure was found in a mixture with an oxyfluoride that had a fluorite-like structure. Since (*2*) observed a transition in YbF_3 with absence of any extraneous phase it can be suggested that it was possible in that case to keep pyrohydrolysis to a minimum. The mechanism of the oxygen effect on the temperature of the polymorphic transformations of dimorphic LnF_3 from ErF_3 to LuF_3 and YF_3 requires further study. However, preliminary results (see Fig. 5, curve 3) indicate that the stabilizing effect of replacement of fluorine by oxygen in the α-YF_3 structural type increases with increase in the atomic number of the lanthanide.

Let us now look at the data in the literature on the fusion and polymorphic transition temperatures of LnF_3 in the light of the experimental data revealed above.

As was said previously, the greatest scattering of transition temperatures is noticed in "GdF_3." (The formula is put in quotation marks since no change in chemical composition is reflected in it as a result of thermal processing). However, from Table I it can be seen that the temperatures obtained are grouped around two values: 900 and ~1070°C. The former temperature coincides well with the temperature we have determined for the eutectoid dissolution of $GdF_{3-2x}O_x$ (888°C) and the latter, with the transition to rather pure GdF_3. Not long ago, (*13*) presented information on the lowering of the transition temperature in "GdF_3" from 1100 to 900°C on its second heating in an open crucible in a stream of helium; (*6*) observed a decrease in the transition temperature from 1060 to 1033°C

on adding Gd_2O_3 to GdF_3. Another conclusion that follows from a juxtaposition of our data with (*10*, *11*) is that "GdF_3" is dimorphic when the oxygen content is from 0.008 to 0.040 wt%.

Our experimental data (Fig. 3C) explain very well the temperature changes of the phase transitions in "HoF_3", which had been heated by (*13*) in an open crucible. In the first heating (*13*), we observed transformations at 1145 and 1125°C; in the second heating, a new effect occurred at 1075°C. According to our data, the temperature of the eutectic between HoF_3 and $HoF_{3-2x}O_x$ is 1128°C, and temperature of the incongruent fusion of $HoF_{3-2x}O_x$ is 1147°C. With further hydrolysis an effect should appear from the eutectoid decomposition of the $HoF_{3-2x}O_x$ phase, which, according to our measurements, occurs at 1082°C.

Thus, in the first part of the present work the effect of admixture of oxygen on the thermal characteristics of LnF_3 has been explained. In order to give a complete picture of the phase transformations in the lanthanide trifluorides, it is necessary to define more accurately not only the temperatures of these transformations but also the limits of the existence of the various structural types of LnF_3. The following article is devoted basically to this analysis.

Acknowledgment

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